

CHARLOTTETOWN, PE, Canada

WMO: 717060

Lat: 46.289N

Lon: 63.129W

Elev: 159

StdP: 14.61

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.7	1.3	-13.7	2.6	-1.8	-8.9	3.4	2.4	33.5	31.7	29.9	30.3	10.4	290	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	80.6	69.5	78.0	67.7	75.6	66.3	71.5	77.7	69.9	75.5	68.2	73.5	11.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.4	108.6	75.0	67.7	102.5	73.4	66.2	97.0	71.8	35.4	77.8	34.0	75.6	32.7	73.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.0	23.3	20.7	DB	-8.4	85.7	4.4	2.3	-11.6	87.3	-14.2	88.7	-16.7	90.0	-19.9	91.7	(4)
(5)			WB	-9.2	74.5	4.3	1.5	-12.3	75.6	-14.9	76.5	-17.3	77.3	-20.5	78.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.3	20.0	20.3	27.5	37.9	48.7	58.4	66.8	65.9	58.8	48.3	38.1	27.8	(6)	
	DBStd	18.00	10.50	9.68	8.25	7.01	7.01	6.91	4.94	5.06	6.47	6.83	8.06	8.88	(7)	
	HDD50	4127	929	832	697	371	111	10	0	0	6	117	367	687	(8)	
	HDD65	8119	1394	1252	1161	814	506	216	40	51	205	520	808	1152	(9)	
	CDD50	1697	0	0	1	7	71	263	520	494	270	62	9	0	(10)	
	CDD65	214	0	0	0	0	1	19	95	80	18	1	0	0	(11)	
	CDH74	872	0	0	0	0	11	93	392	316	59	1	0	0	(12)	
	CDH80	96	0	0	0	0	0	6	42	41	7	0	0	0	(13)	
(14) Wind	WSAvg	10.2	11.2	11.0	11.3	10.9	10.1	9.4	9.0	8.5	9.2	10.1	10.9	11.5	(14)	
(15) Precipitation	PrecAvg	47.9	4.4	3.7	3.9	3.6	3.7	3.7	3.3	3.5	3.8	4.4	4.7	5.2	(15)	
	PrecMax	62.2	10.4	6.9	7.5	7.0	8.6	8.0	9.1	8.3	7.0	8.5	8.5	10.2	(16)	
	PrecMin	33.4	1.3	1.1	1.8	0.6	1.0	0.8	0.4	0.4	0.9	1.1	1.6	2.5	(17)	
	PrecStd	6.8	1.6	1.4	1.3	1.4	1.7	1.7	1.9	1.9	1.5	1.9	1.5	1.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.8	47.2	52.4	64.0	75.5	80.3	84.1	84.2	79.6	70.4	62.3	53.5	(19)	
		MCWB	49.7	45.7	45.7	54.8	61.1	68.0	71.6	70.6	69.7	62.5	59.3	52.3	(20)	
	2%	DB	43.7	41.7	46.1	57.7	70.1	76.7	80.8	80.4	75.1	65.4	58.4	48.5	(21)	
		MCWB	42.9	40.0	43.0	51.0	58.7	65.6	70.0	69.2	67.0	60.2	56.3	47.5	(22)	
	5%	DB	38.9	37.5	42.2	53.2	65.1	73.6	78.4	77.4	71.8	62.1	54.9	44.0	(23)	
		MCWB	37.3	35.6	39.1	48.2	55.6	63.7	68.2	67.5	65.0	58.2	52.7	42.5	(24)	
	10%	DB	34.4	34.2	38.7	49.2	60.9	70.2	75.8	74.9	68.9	59.0	50.5	39.7	(25)	
		MCWB	33.1	32.4	35.9	44.3	53.6	61.9	66.8	66.1	63.4	55.1	48.0	37.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.6	46.2	48.1	58.2	64.1	71.2	73.9	73.6	71.1	64.8	60.1	52.7	(27)	
		MCDB	50.6	46.7	50.3	61.5	70.7	78.3	81.1	80.1	76.5	67.9	61.7	53.4	(28)	
	2%	WB	43.2	40.3	43.7	53.3	60.6	68.1	71.9	71.3	68.8	61.6	56.8	47.8	(29)	
		MCDB	43.8	41.4	45.6	56.3	66.8	74.0	78.6	77.1	73.0	64.4	58.3	48.5	(30)	
	5%	WB	37.5	35.7	39.8	49.1	57.6	65.6	70.1	69.5	66.7	58.8	53.0	42.9	(31)	
		MCDB	38.5	37.0	41.9	52.2	63.0	71.1	75.7	75.1	70.5	61.7	54.6	43.8	(32)	
	10%	WB	33.2	32.4	36.1	44.7	54.5	63.1	68.4	67.8	64.4	55.7	48.3	38.3	(33)	
		MCDB	34.2	33.8	38.3	48.6	59.8	68.5	73.9	72.9	67.9	58.6	50.2	39.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	13.9	12.6	13.7	16.6	16.1	14.7	14.5	14.3	12.5	11.3	11.2	(35)	
		MCDBR	20.6	16.8	16.7	20.5	23.8	20.4	18.0	17.5	16.8	15.6	16.0	16.6	(36)	
	5% WB	MCWBR	21.3	16.5	14.8	14.9	14.5	12.2	9.8	9.2	10.7	12.1	15.4	17.5	(37)	
		MCWBR	20.5	17.1	16.2	18.3	20.3	17.8	15.2	15.0	14.9	14.7	15.9	16.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.299	0.289	0.318	0.348	0.365	0.381	0.407	0.393	0.355	0.333	0.312	0.289	(40)	
		taud	2.343	2.306	2.321	2.396	2.415	2.388	2.353	2.387	2.491	2.528	2.505	2.465	(41)	
	Ebn at Noon	Ebn at Noon	251	281	288	289	286	280	271	271	273	264	248	246	(42)	
		Edh at Noon	24	30	34	35	36	37	38	35	29	24	21	19	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	343	568	998	1328	1613	1754	1785	1577	1207	746	387	263	(44)	
	RadStd	RadStd	23	75	93	121	149	134	120	100	105	77	47	23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	+0.94	+1.28	N/A	N/A	N/A	+34
Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.63	N/A	N/A	N/A	N/A

Nomenclature: See separate page